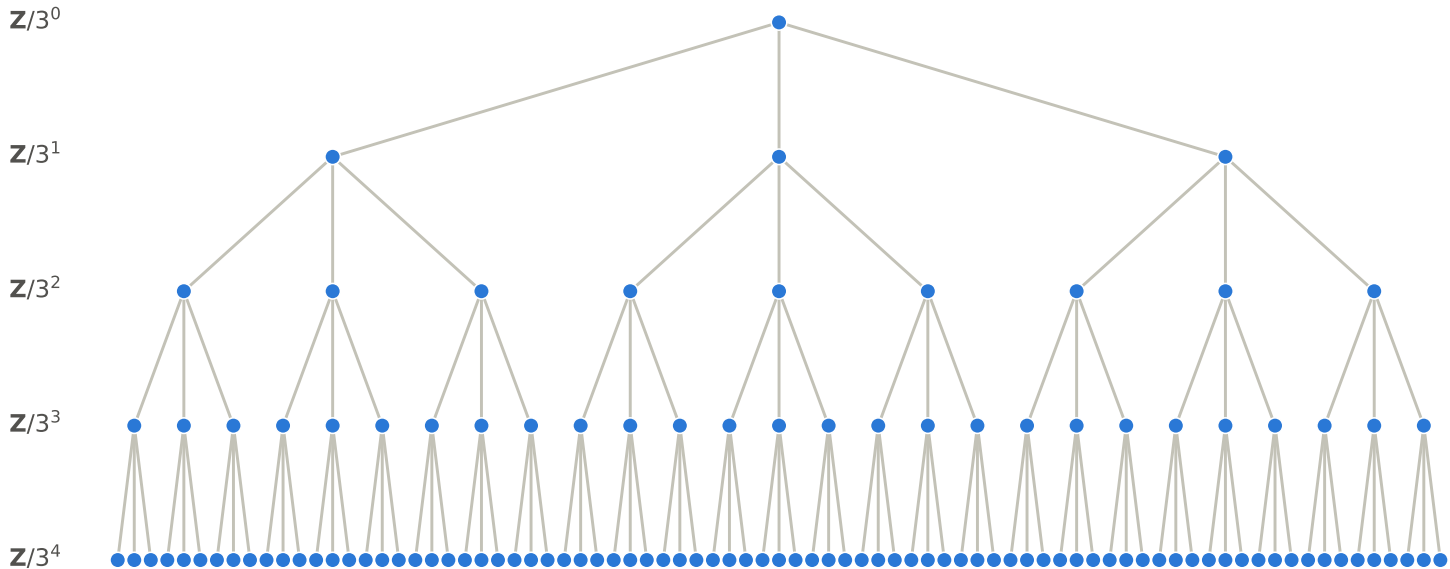

$\mathbf{Z}_3$ is the inverse limit of the $\mathbf{Z}/3^n$: every branch is a 3-adic integer.